

Build inspect adapt

AICE3002 / AICE6001

VLC = $\int \int \int$ Vision
Learning α VoLaC
Control

Convolutional Neural Networks III

Implementation, diagnosis, features, and transfer

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Turning an architecture diagram into code

Tracking tensor shapes through a CNN

Checking parameter counts and receptive fields

Feature hierarchies emerge across depth

Visualising activations and filters

Diagnosing optimisation and generalisation failures

What a pretrained representation provides

Feature extraction versus fine-tuning

Designing a controlled architecture comparison

Separating architectural gains from training gains

A practical CNN investigation